

ANG-Hypergraph v1.0 — Pure-Ontology Local Computational Specification

0. Metadata

- **Version:** v1.0-Stable
- **System Affiliation:** ANG-TOE / TSAS Full-Discipline Unified Framework
- **Core Paradigm:** Local Sub-Hypergraph Conservation + Global Axiom Zero Constraint
- **Default Runtime Mode:** Pure-Ontology (Projection OFF, human perception layer disabled)
- **Computation Property:** Fully-local, sparse, adaptive, no global universe traversal
- **License:** CC-BY-SA 4.0
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- **Archive Repository:** DOI: <https://doi.org/10.5281/zenodo.21660538/> GitHub: https://github.com/ChengbinSong/UVMM_ANG_TOE-Unified-Vacuum-Medium-Model_Angular-Momentum-Network-Geometry
- **Document Status:** Official Stable Specification
- **Last Updated:** 2026-08-19
- **Compatible Frameworks:** ANG-TOE Pairwise Graph / TSAS Continuous Geometry / ANG-Hypergraph

1. Abstract

This document defines the **ANG-Hypergraph** discrete computational kernel, the optimal discrete implementation of the Axiom-0 zero-global-angular-momentum ontology. Unlike ordinary pairwise graphs that only support two-body interaction, the hypergraph structure enables native multi-body topological coupling and one-step multi-node angular-momentum redistribution.

The framework strictly separates **global axiom constraint** and **local subdomain computation**. Local simulations only enforce boundary flux balance instead of zero local total momentum, eliminating redundant full-universe scanning and global correction overhead. The main evolution loop runs entirely in native 6D ontology

space; 3D human-perception projection is downgraded to an on-demand snapshot probe.

2. Three Core Computational Frameworks (Full System Baseline)

This document fully integrates the three native computational frameworks of the ANG-TOE/TSAS unified ontology system. All three frameworks follow the universal Axiom-0 (global zero angular-momentum constraint) but adopt different mathematical carriers, applicable to theoretical proof, high-precision continuous calculation, and sparse discrete local simulation respectively. AI can directly identify and switch three computing paradigms through this chapter.

2.1 ANG-TOE Original Framework (Pairwise Graph Baseline)

2.1.1 Core Definition

The original ANG-TOE system is built on **ordinary pairwise undirected graph**, the earliest discrete implementation of the zero angular-momentum ontology:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E})$$

- **$\mathcal{V}$ Vertex** : Fundamental 6D angular-momentum unit $\mathbf{J}_v \in \mathbb{R}^6$, consistent with hypergraph vertex definition.
- **$\mathcal{E}$ Pairwise Edge** : Each edge only couples two vertices, realizing two-body angular-momentum exchange and propagation.

2.1.2 Core Paradigm

- Global constraint: Closed universe graph satisfies $\sum \mathbf{J}_v = \mathbf{0}$
- Local evolution: All dynamic interactions are decomposed into pairwise neighbor-to-neighbor momentum transmission
- Operation mode: Local sparse iteration, no mandatory global traversal in optimized implementation

2.1.3 Advantages & Limitations

- **Advantages**: Intuitive logic, extremely low development threshold, perfect for rapid prototype verification and small-scale local simulation.
- **Limitations**: Cannot natively support multi-body topological coupling; complex multi-node phase transitions require massive iterative decomposition, low computational efficiency for large-scale topological evolution.

2.2 TSAS Continuous Geometric Framework (High-Precision Analytical Baseline)

2.2.1 Core Definition

TSAS (Topological-Superfluid Axiom System) is the **continuous differential geometric core** of the unified ontology, based on symplectic manifold reduction and $SU(3)$ gauge geometry, abandoning discrete graph carriers:

- **Ontology Carrier:** 6D symplectic manifold $\mathcal{M}_6$, reduced manifold $\mathcal{M}_6^{\text{red}} = \mu^{-1}(0)/G$
- **Core Geometric Objects:** Connection 1-form $\mathbb{A}$, curvature 2-form $\mathbb{F}$, 6D Dirac operator, Hopf invariant $\mathcal{H}$, Pontryagin class p_1
- **Core Constraint:** Moment-map zero constraint $\mathbf{J} = 0$, intrinsically embedding the global zero angular-momentum axiom into manifold topology

2.2.2 Core Paradigm

- All physical disciplines and topological phenomena are projection products of high-dimensional geometric structures
- No manual constraint correction: The reduced manifold only contains state sets that naturally satisfy Axiom-0
- Topological judgment relies on intrinsic invariants: Spectral gap Δ_{gap} , topological transition cost ΔCS
- Default pure-ontology calculation, 3D perception projection is an on-demand external output

2.2.3 Advantages & Limitations

- **Advantages:** Extreme high precision, unified full-discipline analytical equations, exponential convergence spectral solving, accurate topological phase transition calibration.
- **Limitations:** High initialization cost of geometric operators, poor adaptability for local sparse small-scale dynamic simulation, unsuitable for rapid iterative evolution of discrete local domains.

2.3 ANG-Hypergraph Framework (Optimized Discrete Sparse Baseline)

2.3.1 Core Definition

The upgraded discrete computing framework in this document, serving as the **optimal local computational engine** matching TSAS continuous geometry:

$$\mathcal{H} = (\mathcal{V}, \mathcal{E}_h, \Phi)$$

- $\mathcal{V}$ **Vertex** : Consistent with ANG-TOE 6D angular-momentum unit, maintaining framework compatibility
- $\mathcal{E}_h$ **Hyper-edge** : Break pairwise restriction, support arbitrary multi-body synchronous coupling
- Φ **Boundary Flux** : Unified discrete boundary condition matching TSAS Robin boundary geometry

2.3.2 Core Paradigm

- Global axiom restraint + local flux balance separation design
- Multi-body topological interaction one-step calculation, no pairwise decomposition redundancy
- Adaptive sparse dormant domain unloading, zero invalid calculation overhead
- Pure-ontology evolution by default, compatible with TSAS invariant topological detection

2.3.3 Advantages & Limitations

- **Advantages:** Inherit ANG-TOE local sparse advantage + TSAS topological intrinsic characteristics, fastest discrete local simulation, perfect matching of continuous-discrete dual-layer mapping.
- **Limitations:** Partial hypergraph differential operators require independent implementation, relying on TSAS for high-precision critical point calibration.

2.4 Three-Framework Unified Core Consistency

All three frameworks are completely isomorphic in ontology, only different in mathematical computing carriers:

1. Follow the only core axiom: **Global closed system zero total angular momentum**
2. Unified runtime rule: **Local subdomain does not require zero momentum, only flux balance**
3. Unified operating mode: **Pure-ontology first, 3D projection on-demand probe only**
4. Unified topological criterion: Local structural mutation corresponds to system stability failure

2.5 Three-Framework Scenario Partitioning

Framework	Core Application Scenario	Computing Characteristics
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ANG-TOE Pairwise Graph	Prototype verification, small-scale simple dynamic simulation, beginner docking	Fast startup, low threshold, low efficiency for complex topology
TSAS Continuous Geometry	Theoretical proof, phase transition calibration, high-precision quantitative prediction, full-discipline unified equation derivation	High precision, spectral exponential convergence, high initialization cost
ANG-Hypergraph	Large-scale local sparse simulation, multi-body topological evolution, daily main computing engine	Zero global overhead, adaptive sparsity, compatible with both discrete and continuous layers

2.6 Global Axiom-0 (Invariant Meta Constraint)

Unified meta axiom for all three frameworks, no iteration correction, only ontology restraint:

$$\sum_{v \in \mathcal{H}_U / \mathcal{G}_U} J_v = \mathbf{0}$$

This axiom acts on the complete closed universe system and is the only fundamental constraint of the entire unified ontology system.

2.7 Local Subdomain Unified Conservation Rule

All local open subdomain simulations of the three frameworks follow the same flux balance equation:

$$\frac{d}{dt} \Sigma J_{\text{local}} = -\Sigma \Phi_{\text{boundary}}$$

Local total momentum is dynamically floating, and all local calculations eliminate redundant global full-domain traversal and correction.

2.8 Hypergraph Ontology Structure

The ANG-Hypergraph is formally defined as a triple:

$$\mathcal{H} = (\mathcal{V}, \mathcal{E}_h, \Phi)$$

- **$\mathcal{V}$ Vertex set:** Each vertex $v \in \mathcal{V}$ represents a fundamental 6D angular-momentum unit, carrying intrinsic state vector $J_v \in \mathbb{R}^6$.

- **$\mathcal{E}_h$ Hyper-edge set** : A single hyper-edge $e_h \in \mathcal{E}_h$ couples arbitrary $k \geq 2$ vertices, supporting synchronous multi-body topological interaction without decomposing into pairwise iterations.
- **Φ Boundary flux** : Hyper-edge flux across subdomain boundaries, governing all open-system momentum exchange.

3. Key Advantages Over Ordinary Graphs

- **Native multi-body topological coupling**: One hyperedge completes multi-node momentum synchronization in a single step, eliminating redundant intermediate iterations required by pairwise graph decomposition.
- **No global universe traversal**: Local simulation only evolves active sub-hypergraph; dormant regions are evicted from memory, achieving full sparse adaptive computation.
- **Unified boundary topology**: Boundary hyper-edges integrate discrete flux boundary conditions, perfectly matching the Robin boundary rules of continuous TSAS/Lie-Poisson geometry.
- **Pure-ontology first**: Evolution loop operates purely on 6D angular-momentum ontology; 3D projection is optional and non-blocking.

4. Runtime Execution Modes

4.1 Default Mode: Pure-Ontology (Projection = OFF)

All core dynamics run in native 6D ontology space. No 3D spacetime, mass, temperature or other human-perceptible observables are computed during iteration. Topological state and stability are judged by intrinsic hypergraph structure and flux variation.

4.2 On-Demand Mode: Probe Snapshot (Projection = ONE-SHOT)

Projection operator $\hat{\Pi}$ is only invoked when human-readable observation data is required. After snapshot output, the system immediately returns to pure-ontology mode. No persistent projection overhead exists.

5. Standard Evolution Algorithm (Discrete Time Step)

1. **Active Domain Screening**: Filter active hyper-edges with non-zero flux variation; skip all dormant hyper-edges and static vertex regions to save computation.

2. **Intra-Hyperedge Redistribution:** Redistribute angular momentum among all vertices on each active hyperedge, strictly satisfying local intra-edge conservation $\sum \Delta J_v = 0$.
3. **Boundary Flux Update:** Refresh flux values of all boundary hyper-edges to maintain open-system local balance.
4. **Topological Monitoring (Optional):** Attach persistent homology module to detect homology loop mutation and early-warning topological phase transition.
5. **No Global Correction:** No full-graph summation, no global momentum rescaling at any step.
6. **Probe Trigger Check:** Execute one-shot projection if snapshot is required; otherwise loop directly.

6. Bidirectional Mapping: ANG-Hypergraph ↔ Continuous Geometry Kernel

The hypergraph discrete engine is fully compatible with TSAS/Lie-Poisson continuous geometric framework, forming a dual-layer unified system:

ANG-Hypergraph (Discrete Computation Layer)	Continuous Ontology Geometry Layer
Vertex 6D angular-momentum J_v	Local moment-map value on reduced manifold
Internal hyper-edge e_h	Local connection flux tube & geometric coupling channel
Boundary hyper-edge flux Φ	Robin boundary condition of symplectic/Lie-Poisson manifold
Hypergraph loop structure	Homotopy loop / Hopf topological invariant
Topology critical signal	Spectral gap $\Delta_{\text{gap}} \rightarrow 0$

7. Executable Runtime Pseudocode

```
python
# ANG-Hypergraph v1.0 Pure-Ontology Runtime Core
INIT LOCAL_HYPERGRAPH = (V_active, E_h_active, boundary_flux_phi)
PROJECTION_ENABLED = False # Default pure ontology mode
```

```

while simulation_running:
    # Step 1: Adaptive sparse update (skip dormant structure)
    active_hyperedges = filter_active(E_h_active)

    # Step 2: Local multi-body momentum redistribution
    for eh in active_hyperedges:
        vertex_group = eh.linked_vertices
        redistribute_angular_momentum(vertex_group)
        # Strict intra-edge local conservation
        assert sum_delta_J(vertex_group) == 0

    # Step 3: Update open boundary flux
    update_boundary_flux(boundary_flux_phi)

    # Step 4: Lightweight topological detection
    topology_risk = persistent_homology_detect(LOCAL_HYPERGRAPH)
    if topology_risk == CRITICAL:
        switch_high_precision_continuous_solver()

    # Step 5: On-demand probe projection only
    if probe_snapshot_flag:
        ontology_state = export_hypergraph_state()
        observable_3d = projection_operator(ontology_state)
        save_snapshot(observable_3d)
        probe_snapshot_flag = False

# End of pure-ontology main loop

```

8. System Architecture Positioning

- **Discrete Layer (Main Computational Engine):** ANG-Hypergraph — sparse, local, fast adaptive simulation for all regional dynamic evolution.
- **Continuous Layer (High-Precision Calibration):** TSAS Symplectic / Lie-Poisson Manifold — analytical proof, topological transition calibration, invariant verification.
- **Observation Layer (Optional Output):** 3D perception slice projection — only for human observation and experimental data fitting.

9. Version Revision Log

- **v1.0-Stable:** Official release of ANG-Hypergraph specification. Implements pure-ontology local computation, eliminates global full-graph correction, establishes dual mapping with continuous geometric kernel, supports multi-body hyperedge coupling and adaptive sparse iteration.

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